Credit-Risk: Loan Default Prediction

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Abstract

Credit card lenders must decide which clients to grant credit to without knowing who will repay and poor decisions can lead to serious losses for both lenders and borrowers. This study compared logistic regression, random forest and XGBoost for predicting whether a credit card client will default on their next payment and showed how the predictions can be turned into practical lending decisions. The models were trained on the Default of Credit Card Clients dataset, which contains 29,965 clients of a Taiwanese bank observed between April and September 2005, after cleaning. Three additional features describing credit utilisation, repayment history and payment behaviour were constructed, class imbalance was handled with class weights and hyperparameters were tuned by five-fold cross-validation. The models were evaluated on a separate test set using the AUC, Gini coefficient, Kolmogorov–Smirnov statistic, Brier score, precision and recall. XGBoost performed best, with an AUC of 0.783 and a Gini coefficient of 0.565, compared with 0.778 for the random forest and 0.752 for logistic regression. The number of months with a late payment and the most recent repayment status were the strongest predictors of default. After correcting for class weighting, the predicted probabilities were well calibrated and a cost-based cut-off of 0.17 allowed XGBoost to identify 74% of defaulters, reduce the default rate among approved clients from 22.1% to 10.1% and halve the cost of approving every applicant. Excluding sex, age and marital status had almost no effect on accuracy and the resulting model treated different groups of clients in a similar way. These results show that machine learning models can support accurate, cost-effective and fair lending decisions.

Keywords: credit risk; probability of default; credit scoring; logistic regression; XGBoost; machine learning

1 Introduction

Credit risk is the possibility that a borrower will fail to repay a loan or meet other contractual obligations, causing the lender to lose part or all of the principal and the interest owed [1]. It arises in almost every form of lending, including personal loans, mortgages, bonds and credit cards. Credit cards carry particularly high credit risk because they are unsecured: the lender holds no collateral that can be sold if the client stops paying. For this reason, banks need reliable ways to estimate each client's probability of default before granting or extending credit. Accurate estimates allow a lender to approve good clients, decline or limit risky ones and set aside enough capital to cover expected losses [1].

The consequences of poor credit risk assessment were seen in Taiwan during 2005 and 2006, when card-issuing banks expanded credit too quickly to clients who could not afford to repay [2]. The resulting rise in defaults caused serious losses for the banks and left many cardholders in heavy debt. The dataset used in this study comes from a Taiwanese bank during this period, which makes it a realistic setting in which to study default prediction.

Credit scoring treats default prediction as a classification problem: using information about a client, the lender estimates whether the client will repay or default. Hand and Henley [3] reviewed the statistical methods used for this purpose, including discriminant analysis, logistic regression, decision trees, nearest-neighbour methods and neural networks. They noted that no single method is best for every dataset and that differences in accuracy between well-applied methods are often small. Logistic regression became the industry standard [1] because it estimates the probability of default directly and its coefficients can be read as changes in the odds of default. This interpretability matters because lenders must be able to explain their decisions to clients and regulators.

As computing power has grown, many machine learning methods have been applied to credit scoring. Baesens et al. [4] compared a wide range of classification algorithms on several real credit datasets and found that, although support vector machines and neural networks performed well, simple methods such as logistic regression and linear discriminant analysis were also highly competitive. Lessmann et al. [5] updated this benchmark with a much larger set of classifiers, including ensemble methods that combine many models. They found that ensembles generally outperformed individual classifiers and they recommended that random forests replace logistic regression as the standard benchmark in credit scoring research.

Random forests [6] build many decision trees on bootstrap samples of the data, each using a random subset of the variables at every split and average their predictions. Gradient boosting builds trees one after another, with each new tree correcting the errors of the previous ones; XGBoost [7] is an efficient and regularised implementation of gradient boosting that is now widely used for tabular data. Both methods can capture non-linear relationships and interactions between variables that a linear model may miss. However, they are harder to interpret than logistic regression, so the gain in accuracy must be weighed against the loss of transparency.

Yeh and Lien [2] introduced the dataset used in this study and compared six data mining techniques: logistic regression, discriminant analysis, naive Bayes, k-nearest neighbours, classification trees and artificial neural networks. They argued that the estimated probability of default is more useful for risk management than a simple classification of clients as credible or not credible. Because the true probability of default cannot be observed, they proposed a Sorting Smoothing Method to estimate it and compared it with the probability predicted by each

model. The artificial neural network was the only technique whose predicted probabilities closely matched the estimated real probabilities. The dataset has since become a common benchmark for credit scoring methods.

A model that predicts the probability of default does not by itself tell a bank which applications to approve. The bank must choose a cut-off probability and this choice depends on the costs of the two possible errors: approving a client who later defaults and rejecting a client who would have repaid. Elkan [8] showed that, when these costs are known, the cost-minimising cut-off follows directly from the ratio of the two costs, rather than the default value of 0.5. Such decisions also require the predicted probabilities to be well calibrated, meaning that clients given a probability of, say, 30% should default about 30% of the time. Niculescu-Mizil and Caruana [9] showed that some powerful classifiers, including boosted trees, can rank cases well but produce poorly calibrated probabilities, which can be corrected using methods such as Platt scaling or isotonic regression.

Credit scoring models can affect people's access to finance, so they should not disadvantage clients on the basis of personal characteristics such as sex, age or marital status. Kozodoi et al. [10] examined several ways of measuring and improving fairness in credit scoring and showed that fairness can often be improved at a modest cost to the lender's profit. Since the Taiwanese dataset contains sex, age and marital status, it provides an opportunity to examine how much these variables contribute to model performance.

2 Problem statement

Credit card lenders must decide which clients to grant credit to without knowing who will repay and poor decisions, as in Taiwan in 2005 and 2006 [2], lead to serious losses for both lenders and borrowers. Although many models have been proposed for predicting default, the original study of the Taiwanese credit card dataset [2] did not consider ensemble methods such as random forests and XGBoost, which later benchmark studies found to be among the strongest classifiers for credit scoring [5]. Moreover, model comparisons rarely ask whether a gain in accuracy justifies the loss of interpretability that lenders and regulators require and they seldom show how predicted probabilities would be turned into approval decisions using the costs of each type of error [8], or whether those probabilities are well calibrated [9]. The use of sex, age and marital status as predictors also raises concerns about fairness that are rarely examined [10]. This study therefore addresses the problem of building a default prediction model that is accurate, interpretable, well calibrated, fair and directly usable for cost-effective lending decisions.

2.1 Aim

The aim of this study is to predict which credit card clients are likely to miss their next payment and to turn these predictions into a practical lending decision.

2.2 Objectives

The specific objectives are to:

1. clean and explore the Default of Credit Card Clients dataset and identify the client characteristics most strongly associated with default;
2. build logistic regression, random forest and XGBoost models to estimate each client's probability of default;
3. compare the models using measures common in credit risk, namely the area under the ROC curve (AUC), the Gini coefficient and the Kolmogorov–Smirnov (KS) statistic, together with precision, recall and calibration;
4. choose a cut-off probability that minimises the expected cost of lending decisions; and
5. examine whether the models treat clients fairly with respect to sex, age and marital status.

3 Methodology

This section describes the three models used to estimate the probability of default and the measures used to evaluate the models and turn their predictions into lending decisions. All analysis is carried out in Python using the pandas, NumPy, scikit-learn, XGBoost and matplotlib libraries.

3.1 Data

The study uses the Default of Credit Card Clients dataset [2], obtained from Kaggle,[1] which contains 30,000 clients of a Taiwanese bank observed from April to September 2005. Each client has 23 explanatory variables (credit limit, sex, education, marital status, age and six months of repayment status, bill amounts and payments) and a binary target indicating default in October 2005; about 22% of clients defaulted. The client ID is removed, duplicates are dropped, undocumented education codes (0, 5, 6) are merged into "other" (4) and marital status 0 into "other" (3). Three features are added: credit utilisation (latest bill / credit limit), number of months late and average payment ratio (amount paid / amount billed).

3.2 Problem formulation

Let $\{(\mathbf{x}_i, y_i)\}_{i=1}^{n}$ denote the data, where $\mathbf{x}_i \in \mathbb{R}^p$ and $y_i \in \{0, 1\}$, with $y_i = 1$ indicating default. Each model estimates

$$p_i = P(y_i = 1 \mid \mathbf{x}_i). \tag{1}$$

The data is split into stratified training (75%) and test (25%) sets. To address class imbalance, defaulters are weighted by the ratio of non-defaulters to defaulters and hyperparameters are chosen by five-fold cross-validation on the training set using the AUC. Since class weighting inflates the predicted probabilities, each predicted probability is corrected by dividing its odds by the class weight, so that the probabilities remain calibrated for the Brier score and the cost-based cut-off.

3.3 Logistic regression

Logistic regression models the log-odds of default as a linear function of the features:

$$\ln\left(\frac{p_i}{1-p_i}\right) = \beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i \quad \Longleftrightarrow \quad p_i = \frac{1}{1 + e^{-(\beta_0 + \boldsymbol{\beta}^\top \mathbf{x}_i)}}. \tag{2}$$

The coefficients maximise the weighted, penalised log-likelihood

$$\ell(\beta_0, \boldsymbol{\beta}) = \sum_{i=1}^{n} w_i \left[y_i \ln p_i + (1 - y_i) \ln(1 - p_i) \right] - \frac{1}{2C} \|\boldsymbol{\beta}\|^2, \tag{3}$$

which is concave, so the solution is unique. The odds ratio e^{β_j} gives the change in the odds of default for a one-unit increase in standardised feature j.

[1] https://www.kaggle.com/datasets/uciml/default-of-credit-card-clients-dataset

3.4 Random forest

A random forest [6] averages B decision trees, each grown on a bootstrap sample with a random subset of features at each split:

$$\hat{p}(\mathbf{x}) = \frac{1}{B}\sum_{b=1}^{B} \hat{p}_b(\mathbf{x}), \tag{4}$$

where splits minimise the Gini impurity $G = 1 - \sum_k \hat{p}_k^2$.

3.5 XGBoost

XGBoost [7] adds trees sequentially, $\hat{z}_i = \sum_{k=1}^{K} f_k(\mathbf{x}_i)$ with $\hat{p}_i = \sigma(\hat{z}_i)$, minimising

$$\mathcal{L} = \sum_{i=1}^{n} l(y_i, \hat{p}_i) + \sum_{k=1}^{K} \left(\gamma T_k + \tfrac{1}{2}\lambda \|\mathbf{w}_k\|^2\right), \tag{5}$$

where l is the log-loss, T_k the number of leaves and $\mathbf{w}_k$ the leaf weights. A second-order Taylor expansion of the loss gives the optimal leaf weight

$$w_j^* = -\frac{G_j}{H_j + \lambda}, \tag{6}$$

where G_j and H_j are the sums of first and second derivatives of the loss in leaf j.

3.6 Evaluation

The models are compared on the test set using

$$\text{Precision} = \frac{TP}{TP + FP}, \qquad \text{Recall} = \frac{TP}{TP + FN}, \qquad \text{Gini} = 2\,\text{AUC} - 1, \tag{7}$$

$$\text{KS} = \max_t |F_1(t) - F_0(t)|, \qquad \text{BS} = \frac{1}{n}\sum_{i=1}^{n} (\hat{p}_i - y_i)^2, \tag{8}$$

where the AUC is the probability that a randomly chosen defaulter receives a higher score than a randomly chosen non-defaulter, F_1 and F_0 are the score distributions of defaulters and non-defaulters and the Brier score (BS) measures calibration [9].

3.7 Cost-based cut-off and fairness

If C_{FN} is the cost of approving a defaulter and C_{FP} the cost of rejecting a good client, the expected cost is minimised by the cut-off [8]

$$t^* = \frac{C_{FP}}{C_{FP} + C_{FN}}. \tag{9}$$

Assuming $C_{FN} : C_{FP} = 5 : 1$ gives $t^* = 1/6 \approx 0.17$; ratios of 3:1 and 10:1 are also tested. Finally, the best model is retrained without sex, age and marital status and approval rates and recall are compared across groups [10]. All analysis is done in Python (pandas, scikit-learn, XGBoost) with a fixed random seed of 42.

4 Results and Discussion

4.1 Data cleaning and exploratory analysis

The raw data contained no missing values. Thirty-five duplicate records were removed, leaving 29,965 clients. A total of 345 clients had undocumented education codes (280 with code 5, 51 with code 6 and 14 with code 0), which were merged into the category "other", increasing it from 123 to 468 clients. Similarly, 54 clients with marital status 0 were merged into "other", increasing it from 323 to 377. After cleaning, 22.1% of clients defaulted.

The most recent repayment status is by far the strongest indicator of default (Figure 1). Clients who used no credit, paid in full or made the minimum payment in September 2005 defaulted at rates of between 12.8% and 16.8%. The rate rose to 34.0% for clients one month late and to 69.1% and 75.8% for clients two and three months late. Fewer than 80 clients were more than three months late, so the rates for these groups are less reliable.

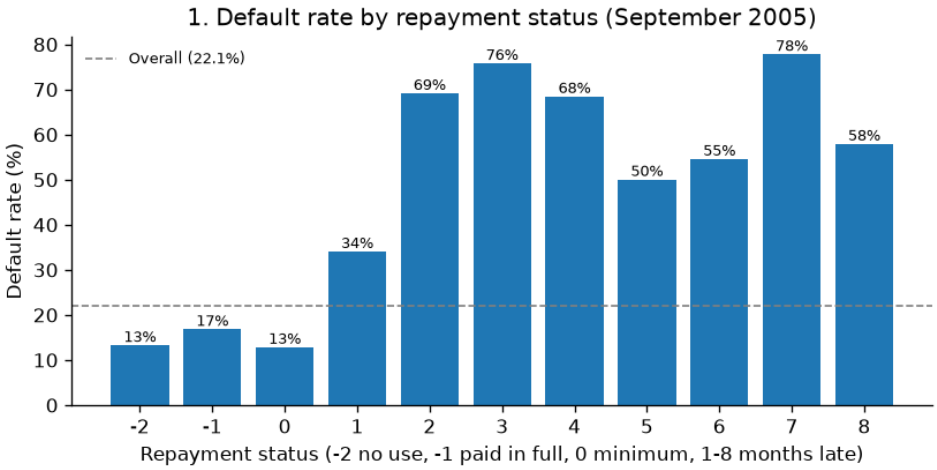


Figure 1: Default rate by repayment status in September 2005.

The default rate also falls steadily as the credit limit increases, from 31.8% for limits of NT$50,000 or less to 11.2% for limits above NT$500,000 (Figure 2), which suggests that the bank already granted higher limits to clients it considered less risky. The relationship with age is U-shaped: the default rate is lowest for clients aged 31 to 35 (19.4%) and highest for the youngest (26.7% for ages 21 to 25) and oldest clients (26.8% for ages over 60).

Differences between demographic groups are smaller. Men defaulted more often than women (24.2% against 20.8%), married clients more often than single clients (23.5% against 20.9%) and the default rate decreased with education, from 25.2% for high school to 19.2% for graduate school. These differences, although modest, show why the fairness of models that use these variables needs to be examined.

The six repayment status variables have the strongest correlations with default, ranging from 0.325 for the most recent month to 0.187 for the earliest, while the credit limit is negatively correlated with default (-0.154). The monthly bill amounts are very strongly correlated with each other (0.951 between consecutive months), since a client's balance largely carries over from one month to the next (Figure 3). This multicollinearity can make individual logistic regression coefficients unstable, which is one reason for including the L_2 penalty in the logistic regression model.

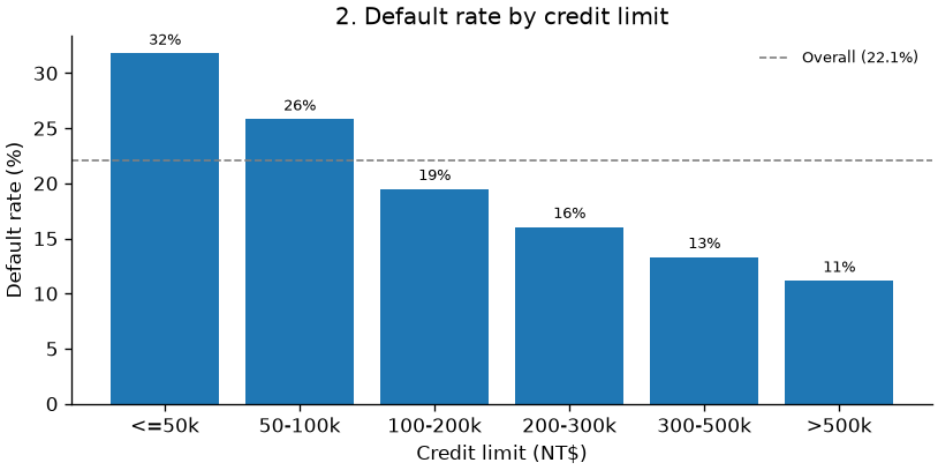


Figure 2: Default rate by credit limit.

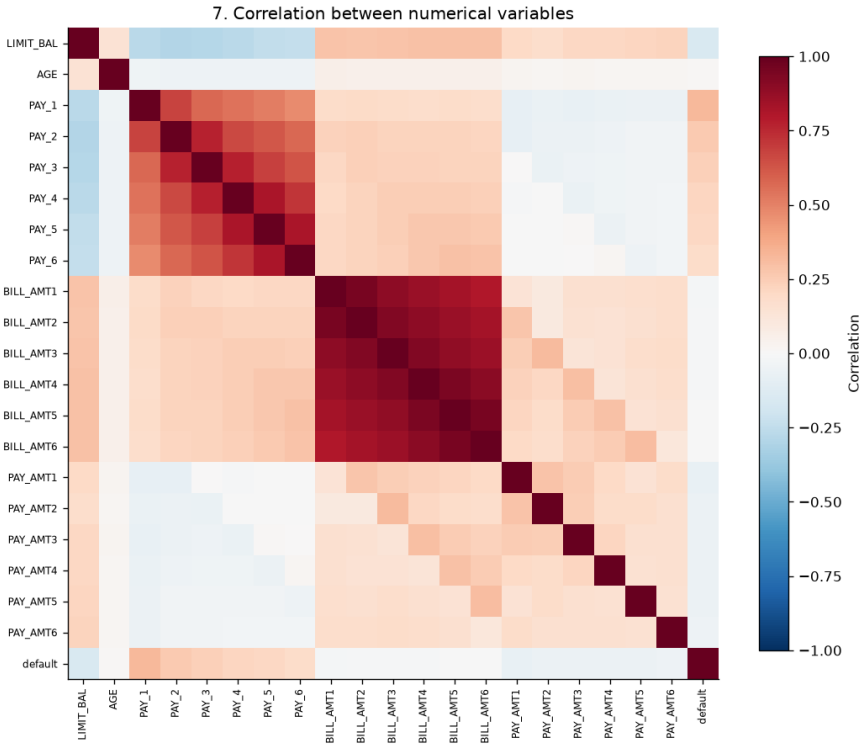


Figure 3: Correlation between the numerical variables and default.

4.2 Model comparison

Table 1 compares the three models on the test set of 7,492 clients. XGBoost achieved the highest AUC (0.783), Gini coefficient (0.565) and KS statistic (0.430), followed closely by the random forest (AUC 0.778), while logistic regression performed worst on every measure (AUC 0.752). The test-set AUCs were very close to the cross-validated AUCs obtained during tuning (0.787 for XGBoost, 0.784 for the random forest and 0.757 for logistic regression), which indicates that the models did not overfit. The ROC curves in Figure 4 show that the two ensemble methods dominate logistic regression across almost all cut-offs. Since XGBoost improved the AUC over logistic regression by 0.031, more than the threshold of 0.02 set in this study, the gain in accuracy is considered large enough to justify the less interpretable model.

Table 1: Performance of the three models on the test set. Precision, recall and cost are at the cut-off $t^* = 0.167$ (cost ratio 5:1).

Model	AUC	Gini	KS	Brier	Precision	Recall	Cost
XGBoost	**0.783**	**0.565**	**0.430**	**0.134**	0.375	**0.744**	**4,178**
Random forest	0.778	0.555	0.422	0.136	**0.406**	0.688	4,254
Logistic regression	0.752	0.504	0.396	0.141	0.396	0.677	4,391

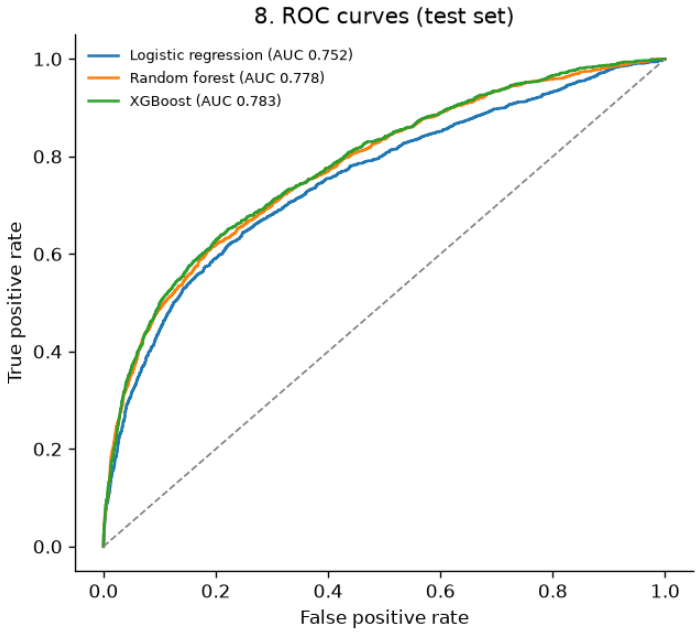


Figure 4: ROC curves of the three models on the test set.

4.3 Calibration and cost-based cut-off

All three models produced well-calibrated probabilities after correcting for class weighting (Figure 5) and XGBoost had the lowest Brier score (0.134). Good calibration is confirmed by the cost analysis: the cut-offs given by Equation (9) were close to the cut-offs that minimised the total cost on the test set, namely 0.25 against 0.26 for a cost ratio of 3:1, 0.17 against 0.14 for 5:1 and 0.09 against 0.09 for 10:1 (Figure 6). Since the empirical minima were found on the test set itself, the formula value $t^* = 0.167$ is used as the main cut-off.

At this cut-off, XGBoost identified 74.4% of defaulters and approved 56.1% of applicants. Only 10.1% of the approved clients defaulted, compared with 22.1% of all clients. The total cost of 4,178 units was about half of the cost of approving every applicant (8,290 units) and 4.9% lower than the cost of using logistic regression.

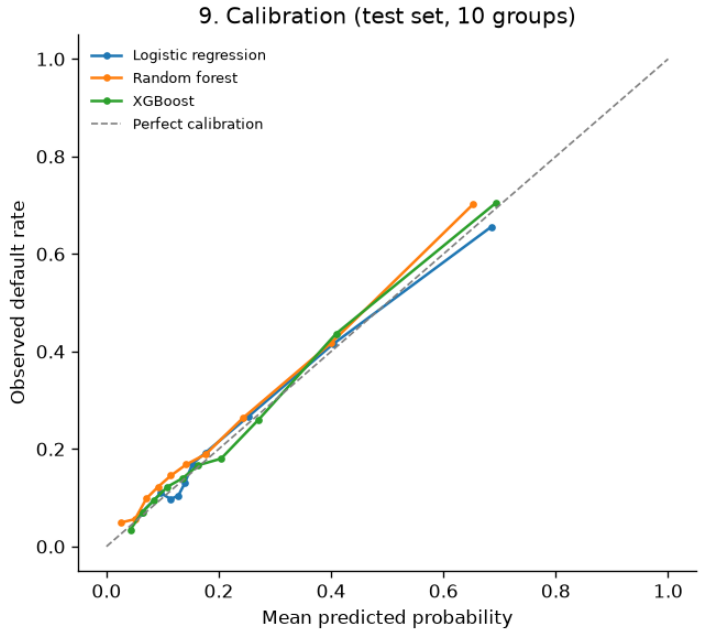


Figure 5: Calibration of the three models on the test set.

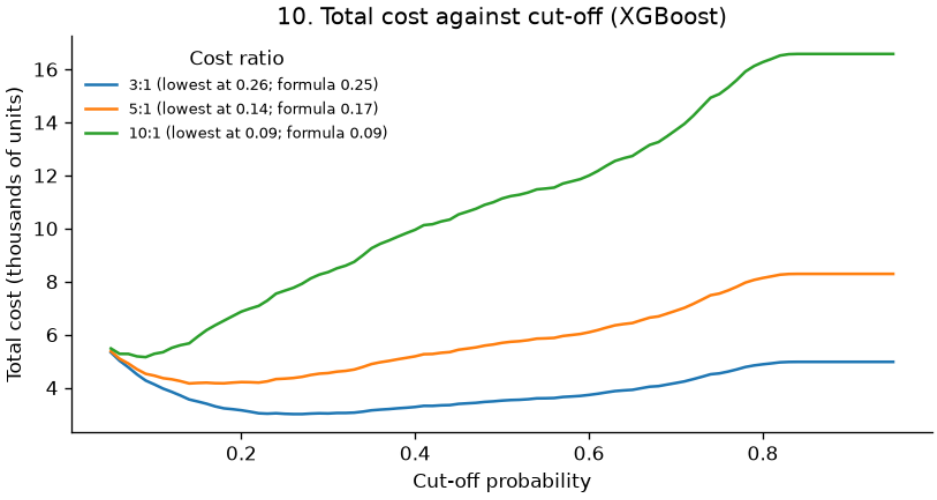


Figure 6: Total cost of XGBoost against the cut-off for cost ratios of 3:1, 5:1 and 10:1.

4.4 Important predictors

Both models identified the number of months in which a payment was late as the strongest predictor of default (Figures 7 and 8). In logistic regression, an increase of one standard deviation in this variable multiplied the odds of default by 2.46 and it accounted for 49.9% of the feature importance in XGBoost. The most recent repayment status was the next most impor-

tant predictor in both models (odds ratio 1.33; importance 12.1%), while a higher credit limit reduced the odds of default (odds ratio 0.78). These results agree with the exploratory analysis. Some bill amounts had odds ratios on opposite sides of one (1.37 for BILL_AMT2 and 0.80 for BILL_AMT1); since these variables are highly correlated, their individual coefficients should not be interpreted separately.

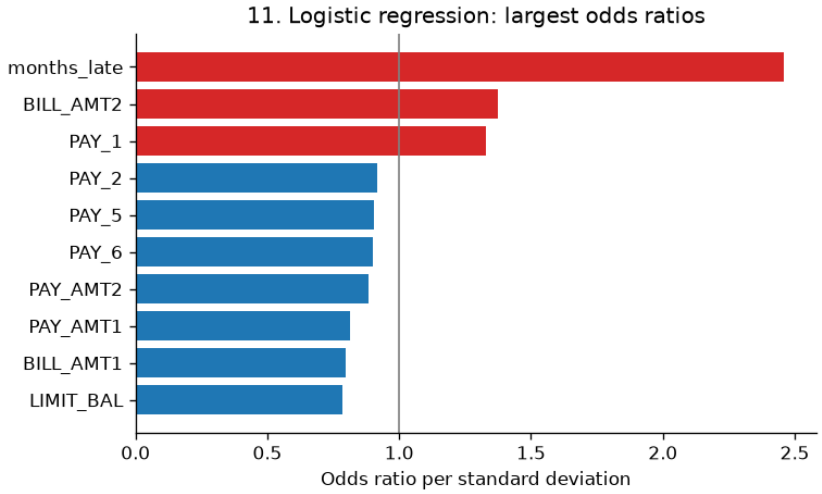


Figure 7: Largest odds ratios of the logistic regression model, per standard deviation.

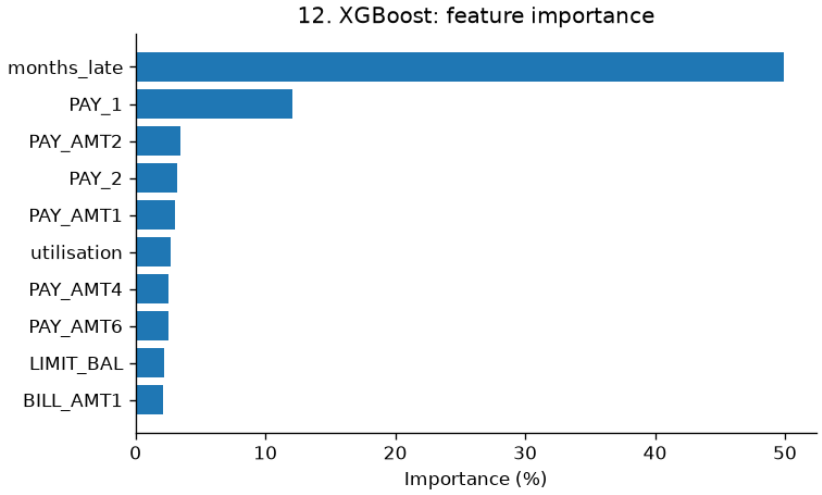


Figure 8: The ten most important features of the XGBoost model.

4.5 Fairness

Removing sex, age and marital status from XGBoost changed the AUC by only −0.0005 (from 0.7827 to 0.7822), so these variables add almost nothing to predictive performance and can be excluded. Without them, the model identified defaulters at similar rates for women and men (recall 0.744 and 0.736) and across age groups (0.717 to 0.764). Women were approved slightly more often than men (57.7% against 54.1%), which is consistent with their lower default rate (20.8% against 24.2%). Approval rates were lowest for the youngest and oldest age groups, which also had higher default rates. The model therefore treats the groups in a similar way and the

differences in approval rates reflect differences in repayment behaviour rather than the use of personal characteristics.

5 Conclusion

This study compared logistic regression, random forest and XGBoost for predicting default among 29,965 credit card clients of a Taiwanese bank and showed how the predictions can be turned into practical lending decisions. XGBoost performed best, with an AUC of 0.783 and a Gini coefficient of 0.565, closely followed by the random forest, while logistic regression achieved an AUC of 0.752. Since the gain of XGBoost over logistic regression exceeded the threshold of 0.02 set in this study, XGBoost is recommended, with odds ratios and feature importance used to explain its main drivers. The number of months with a late payment and the most recent repayment status were the strongest predictors of default in every model, while a higher credit limit was associated with lower risk.

After correcting for class weighting, the predicted probabilities were well calibrated, so the cost-minimising cut-off could be taken directly from the ratio of the error costs. At a cost ratio of 5:1, the corresponding cut-off of 0.17 allowed XGBoost to identify 74% of defaulters, reduced the default rate among approved clients from 22.1% to 10.1% and halved the cost of approving every applicant. Excluding sex, age and marital status had almost no effect on accuracy and the resulting model treated men and women and different age groups in a similar way. A lender could therefore use the model without these personal characteristics at almost no cost.

The study has several limitations. The data comes from a single bank in Taiwan in 2005, before the global financial crisis, so the models demonstrate the method rather than providing a model that could be used directly by lenders in other countries or periods. The costs of each type of error were assumed rather than estimated from real lending data and two tuned hyperparameters lay at the edge of the values considered. Future work could estimate the costs from actual losses and profits, widen the range of hyperparameters, apply the method to more recent data, including South African credit data and use explanation methods such as SHAP values to explain individual predictions of the XGBoost model.

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